Carlo data simulation study

Data sets were generated and models were estimated in R (version 4.5.3) using brms (version 2.23.0) with the cmdstanr backend (CmdStan version 2.39.0). Each model was first estimated with four Markov chain Monte Carlo (MCMC) chains of 8,000 iterations each, of which the first 2,000 were warm-up, resulting in 24,000 post-warm-up draws overall. No thinning was applied. All chains were initialized at zero on the unconstrained parameter scale, and sampling used adapt_delta = 0.98 and max_treedepth = 12. Convergence was assessed for all population-level parameters (fixed effects, random-effect standard deviations and random-effect correlations) using the rank-normalized split-$\hat{R}$ and the bulk and tail effective sample sizes (ESS; Vehtari et al., 2021). A fit was considered converged if the maximum $\hat{R}$ was ≤ 1.01, the minimum bulk and tail ESS were each ≥ 400, and no more than five divergent transitions occurred. Replications that did not meet these criteria were refitted using a fixed sequence of increasingly conservative settings, each with a new random seed and max_treedepth = 15: (1) 10,000 iterations with 2,000 warm-up (32,000 post-warm-up draws) and adapt_delta = 0.99; (2) 16,000 iterations with 4,000 warm-up (48,000 draws) and adapt_delta = 0.995; and (3) 24,000 iterations with 6,000 warm-up (72,000 draws) and adapt_delta = 0.995. Refitting stopped at the first stage that met all criteria, and that fit was used for estimation; all reported convergence diagnostics refer to this final fit.

Across three scenarios, performance was evaluated for parameter bias, average posterior standard deviation, empirical standard deviation, root mean square error, and 95% credible-interval coverage. Following Collins, Schafer, and Kam (2001), we consider as problematic standardized bias values >0.4 and coverage below 90%.

To specify a decaying effect, λ was constrained during model estimation to have a lower bound of 0.

IE Scenario 1: Misspecification of a fixed initial elevation at first occasion as absent

Following the initial fit, 28 replications under the correctly specified model and 12 under the incorrectly specified model did not meet the convergence criteria and were refitted. After this first fit, 2 replications under the correctly specified model still did not meet the criteria and were refitted a second time (stage 2); all replications under the misspecified model met the criteria after the first refit. After refitting, all 250 replications under each model met all convergence criteria: the maximum rhat was 1.0098 under both correctly specified and the misspecified model, all bulk and tail ESS were >= 400, and no final fit had any divergent transitions.

Under this scenario, data were generated under a model in which the initial elevation was represented as a fixed effect at the first assessment. As shown in Table 1, the correctly specified model recovered the generating parameters well. Across parameters, standardized bias was small (range = 0.51% to 19.57%), and empirical sampling variability was generally close to the average model-based standard error. Coverage of nominal 95% intervals ranged from 91.6% to 96.0%, indicating generally adequate parameter recovery and interval calibration.

Conversely, omitting the initial elevation from the fitted model produced bias in several parameters. The largest effects were observed for τ_0, the log of the residual standard deviation, which was estimated at -1.37 compared with the true value of -1.5 (standardized bias = 255.79%), the standard deviation of the random scale effect, which was underestimated (0.397 vs. 0.5; standardized bias = -210.44%), and the overall mean β_0, which was overestimated (2.504 vs 2.5; standardized bias = 109.40%). These biases corresponded to losses in coverage: empirical coverage was 21.6% for τ_0 and 57.2% for ϕ_c , compared with 91.6% and 95.6%, respectively, under the correctly specified model. Coverage for the mean response declined from 95.2% to 82.4%.

For this data-generating scenario, the results indicate that when a fixed initial elevation is confined to a single assessment, treating it as absent can distort estimation of the overall mean (overestimated), the fixed within-subject residual standard deviation (overestimated), the degree of between-subject heterogeneity of the level-1 residual standard deviation (underestimated), and the magnitude of the correlation between the random intercept and the random scale (overestimated).

Table 1

IE Scenario 1: Misspecification of a fixed initial elevation at first occasion as absent

Model	Parameter	True Value	Average Estimate	Average SE	SD of estimates	RMSE	Coverage	% standardized bias
Correct	β_0	2.5	2.504	0.050	0.050	0.050	.952	8.423
	β_1	0.5	0.500	0.027	0.028	0.028	.952	-0.242
	ϕ_b	0.837	0.842	0.035	0.035	0.035	.960	15.359
	ϕ_c	0.354	0.354	0.022	0.020	0.020	.968	2.562
	ρ_{bc}	-.2	-.198	0.070	0.068	0.068	.968	3.223
	τ_0	-0.75	-0.750	0.025	0.026	0.026	.924	0.347
Incorrect	β_0	2.5	2.554	0.050	0.050	0.074	.836	108.590
	ϕ_b	0.837	0.840	0.036	0.034	0.035	.960	10.911
	ϕ_c	0.354	0.314	0.021	0.019	0.044	.572	-204.544
	ρ_{bc}	-.2	-.197	0.073	0.071	0.071	.940	4.397
	τ_0	-0.75	-0.685	0.023	0.025	0.070	.216	260.432

Average Estimate = $\sum_1^R \hat{\theta}/R$. Average SE = $\sum_1^R \widehat{SE(\theta)}/R$. SD of estimates = SD($\hat{\theta}$). RMSE = root mean square error. % standardized bias = 100*[($\sum_1^R \hat{\theta}/R - \theta$)/ SD($\hat{\theta}$)]. Maximum MCSE bias was 0.004.

IE Scenario 2: Misspecification of a fixed, exponentially decaying IE as confined to the first occasion

Final maximum rhat value was 1.008667 for correctly specified models and 1.009862 for incorrectly specified models; all bulk and tail ESS were >= 400. One replication for correctly specified models were refitted and 2 replications for incorrectly specified models were refitted.

In the second scenario, data were generated under a model in which IE was a fixed effect that decayed exponentially over time, and fitted models specified IE as either a first-assessment-only effect or the correct exponential decay. As shown in Table 2, the correctly specified model recovered the generating parameters well. Across parameters, absolute standardized bias was small (range = 0.01% to 19.266%), and empirical sampling variability was generally close to the average model-based standard error. Coverage of nominal 95% intervals ranged from 91.2% to 98.8%, indicating generally adequate parameter recovery and interval calibration.

In contrast, specifying IE as an effect that impacted the first assessment alone produced bias in the overall mean, where the mean response was biased upward (2.53 vs. 2.5; standardized bias = 60.280%), and this corresponded to a loss in coverage, from 95.6% under the correctly specified model to 88.8% under the incorrect model. The magnitude of the initial IE effect was biased downward (0.475 vs. 0.5; standardized bias = -90.502%), and this corresponded to the greatest loss in coverage, from 95.2% under the correctly specified model to 83.2% under the incorrect model. Otherwise, the remaining model parameters were fairly well recovered and estimated coverage was satisfactory.

Table 2

IE Scenario 2: Misspecification of a fixed, exponentially decaying IE as confined to the first occasion

Model	Parameter	True Value	Average Estimate	Average SE	SD of estimates	RMSE	Coverage	% standardized bias
Correct	β_0	2.5	2.504	0.050	0.050	0.050	.956	8.577
	β_1	0.5	0.500	0.027	0.028	0.028	.952	-0.0134
	λ	1.15	1.166	0.126	0.098	0.099	.988	16.779
	ϕ_b	0.7	0.711	0.061	0.058	0.059	.960	19.266
	ϕ_c	0.5	0.505	0.062	0.058	0.058	.960	8.747
	ρ_{bc}	-.2	-.201	0.045	0.044	0.044	.952	-1.754
	τ_0	-1.5	-1.501	0.050	0.053	0.053	.912	-2.004
Incorrect	β_0	2.5	2.53	0.050	0.050	0.058	.888	60.280
	β_1	0.5	0.475	0.028	0.028	0.038	.832	-90.502
	ϕ_b	0.7	0.711	0.061	0.058	0.059	.956	19.027
	ϕ_c	0.5	0.492	0.061	0.056	0.057	.972	-13.547
	ρ_{bc}	-.2	-.198	0.045	0.044	0.044	.956	4.005
	τ_0	-1.5	-1.487	0.050	0.053	0.055	.916	24.051

Average Estimate = $\sum_1^R \hat{\theta}/R$. Average SE = $\sum_1^R \widehat{SE(\theta)}/R$. SD of estimates = SD($\hat{\theta}$). RMSE = root mean square error. % standardized bias = 100*[($\sum_1^R \hat{\theta}/R - \theta$)/ SD($\hat{\theta}$)].

IE Scenario 3: Misspecification of a random, exponentially decaying IE as fixed

Final maximum rhat value was 1.00994 for correctly specified models and 1.009886 for incorrectly specified models; all bulk and tail ESS were >= 400. Five replication for correctly specified models were refitted and 11 replications for incorrectly specified models were refitted.

In the third scenario, data were generated under a model in which IE was a random, exponentially decaying effect, and fitted models specified IE as a fixed, or correctly as random, exponentially decaying effect. As shown in Table 3, the correctly specified model recovered the generating parameters well. Across parameters, absolute standardized bias was small (range = 0.087% to 22.296%), and empirical sampling variability was generally close to the average model-based standard error. Coverage of nominal 95% intervals ranged from 92.8% to 99.2%, indicating generally adequate parameter recovery and interval calibration.

In contrast, specifying IE incorrectly as a fixed effect impacted the estimated costandard deviation structure. At the subject level, the estimated standard deviation of the random intercept was biased upward (0.732 vs. 0.7; standardized bias = 53.565%), and this corresponded to a slight, but still acceptable, decrease in coverage, from 96.0% under the correctly specified model to 94.0% under the incorrect model. The estimated standard deviation of the random scale was biased downward (0.434 vs. 0.5; standardized bias = -118.759%), and this corresponded to a decrease in coverage, from 94.4% under the correctly specified model to 80.8% under the incorrect model. At the occasion level, the estimated log residual standard deviation was biased upward (-1.382 vs. -1.5; standardized bias = 260.421%), and this corresponded to a substantial decrease in coverage, from 96.4% under the correctly specified model to 28.4% under the incorrect model.. The fixed initial elevation effect was also biased upward (0.520 vs. 0.5; standardized bias = 43.570%), and this corresponded to a decrease in coverage, from 92.0% under the correctly specified model to 76.8% under the incorrect model.

The remaining model parameters, including the overall mean, fixed decay effect, and correlation between the random intercept and random scale, were well recovered and estimated coverage was satisfactory.

Table 3

IE Scenario 3: Misspecification of a random, exponentially decaying IE as fixed

Model	Parameter	True Value	Average Estimate	Average SE	SD of estimates	RMSE	Coverage	% standardized bias
Correct	β_0	2.5	2.49	0.050	0.048	0.049	92.8	-12.672
	β_1	0.5	0.499	0.043	0.045	0.045	92.0	-2.762
	λ	1.15	1.16	0.108	0.082	0.082	99.2	12.951
	ϕ_b	0.7	0.713	0.062	0.057	0.059	96.0	22.296
	ϕ_{IE}	0.25	0.250	0.043	0.045	0.045	94.4	0.087
	ϕ_c	0.5	0.508	0.064	0.062	0.062	94.4	12.255
	ρ_{bc}	-.2	-.206	0.071	0.072	0.072	94.8	-8.583
	$\rho_{b,bIE}$	.15	.150	0.085	0.089	0.089	93.6	0.136
	$\rho_{bIE,c}$	-.1	-.090	0.108	0.107	0.107	94.8	8.940
	τ_0	-1.5	-1.50	0.051	0.051	0.051	96.4	6.395
Incorrect	β_0	2.5	2.49	0.051	0.049	0.050	92.8	-19.046
	β_1	0.5	0.520	0.031	0.045	0.049	76.8	43.570
	λ	1.15	1.16	0.128	0.088	0.089	99.6	12.807
	ϕ_b	0.7	0.732	0.063	0.059	0.067	94.0	53.565
	ϕ_c	0.5	0.434	0.056	0.055	0.086	80.8	-118.759
	ρ_{bc}	-.2	-.207	0.072	0.070	0.070	94.4	-10.286
	τ_0	-1.5	-1.38	0.048	0.045	0.126	28.4	260.421

Average Estimate = $\sum_1^R \hat{\theta}/R$. Average SE = $\sum_1^R \widehat{SE(\theta)}/R$. SD of estimates = SD($\hat{\theta}$). RMSE = root mean square error. % standardized bias = 100*[($\sum_1^R \hat{\theta}/R - \theta$)/ SD($\hat{\theta}$)].